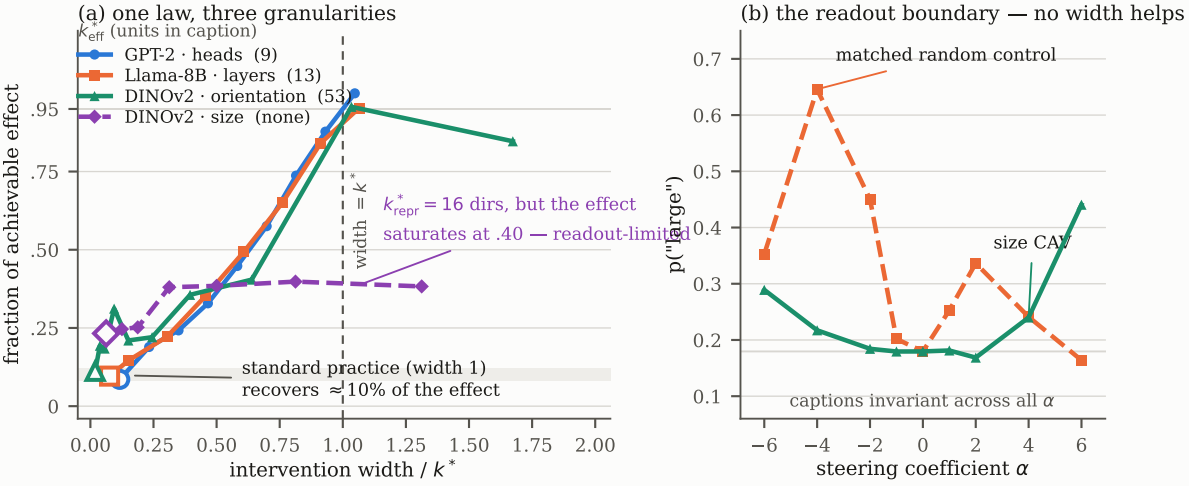
(a) one law, three granularities
k^*_{eff} (units in caption)
GPT-2 · heads (9)
Llama-8B · layers (13)
DINOv2 · orientation (53)
DINOv2 · size (none)
fraction of achievable effect
.95
.75
.50
.25
0
width = k^*
k^*_{repr} = 16 dirs, but the effect
saturates at .40 — readout-limited
standard practice (width 1)
recovers ≈ 10% of the effect
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
intervention width / k^*
(b) the readout boundary — no width helps
matched random control
size CAV
p("large")
0.7
0.6
0.5
0.4
0.3
0.2
0.1
captions invariant across all α
−6
−4
−2
0
2
4
6
steering coefficient α